Journal of Cheminformatics

RESEARCH Open Access

Privileged structure-based molecular fingerprints for organic electronic materials: towards intuitive machine learning interpretation

Tae Wook Yang[1], Seung Hyun Jo[1] and Min Chul Suh[1*]

Abstract

Molecular descriptors are central to the performance and interpretability of QSPR models, yet most existing fingerprints for organic electronics lack chemical relevance or interpretability. Here, we present the Organic Electronic Fingerprint (OEFP), a structure-based representation tailored for OLED and OPV materials. OEFP was constructed from a manually curated OLED dataset and publicly available OPV and chromophore datasets to ensure structural diversity. Synthetically accessible substructures were identified using fragmentation and ring decomposition methods, which capture the conjugated π-bonds crucial for organic electronic materials, and subsequently encoded as individual bits. In case studies of OPV HOMO energy prediction, OEFP achieved up to 13.7% lower MAE and 1.6% higher R^2 compared to domain-mismatched structure-based fingerprint under random splits. Under scaffold-based splits designed to assess generalization, OEFP demonstrated substantially improved generalization, reducing MAE by 50–70% and increasing R^2 by over 150% relative to the same baseline and achieved performance comparable to ECFP baselines evaluated at a similar fingerprint length when using count-based representations. SHAP analysis further enabled intuitive substructure-level interpretation, while OEFP's chemically meaningful and synthetically relevant design supports rational molecular generation. Together, these results establish OEFP as an effective and interpretable molecular representation for machine learning applications in organic electronics.

Scientific Contributions

Benchmarks across diverse datasets demonstrate that the Organic Electronic Fingerprint (OEFP) provides a domain-specific and interpretable molecular representation, capturing chemically meaningful motifs relevant to organic electronic materials. Compared with conventional fingerprints, OEFP consistently improves predictive accuracy and interpretability, while enabling direct substructure–property insights. These results establish OEFP as a valuable resource for advancing machine learning–based molecular property prediction and AI-assisted molecular design in organic electronics.

Keywords Organic electronic materials, Molecular fingerprint, Machine learning, Organic light-emitting diode (OLED), Organic photovoltaics (OPV)

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Introduction

In recent years, machine learning (ML) has emerged as a powerful tool in chemistry and materials science, accelerating workflows in property prediction [1], virtual screening [2], and even inverse molecular design using generative models such as Variational Autoencoders (VAEs) [3, 4]. These advancements are reshaping how researchers navigate chemical space and identify candidates with targeted functionalities. In the domain of organic electronics, ML has gained considerable traction for predicting key molecular properties that influence device performance. Notable efforts have focused on estimating parameters such as highest occupied molecular orbital (HOMO), lowest unoccupied molecular orbital (LUMO) energy levels [1], photoluminescence quantum yield (PLQY) [5], bond dissociation energy (BDE) [6], power conversion efficiency (PCE) [7], and TADF (thermally activated delayed fluorescence) characteristics [2, 8]. These approaches have demonstrated the potential of ML to predict material properties rather than measuring or calculating and support the rational design of high-performance OLED and OPV materials. To enable such predictions, a variety of molecular representations have been explored, including molecular graphs [9], physicochemical descriptors [10], and molecular fingerprints such as the Extended-Connectivity Fingerprint (ECFP) [11]. While these representations enable accurate predictive models, descriptors such as KRFP, which are not domain-specific, or ECFP, which lack explicit structural meaning, fall short in interpretability. As a result, even when the model performs well, it becomes difficult to extract chemically meaningful insights from the output, limiting their utility for rational molecular design. For example, in extended-connectivity fingerprints (ECFP), a single bit may correspond to multiple, structurally distinct substructures depending on the molecular context. Moreover, these substructures are not explicitly defined or synthetically viable, hindering their use in downstream tasks such as rational molecule generation or modification. As a result, it becomes difficult to attribute model predictions to specific chemical motifs. In addition, general-purpose fingerprints [12, 13] tend to overlook domain-specific features such as conjugated cores or donor, acceptor moieties prevalent in organic electronic materials.

Recent high-profile studies on OLED and organic photodetector (OPD) molecules have highlighted the limitations of current molecular representations. For instance, a state-of-the-art model for predicting BDEs in OLEDs employed a VAE-based active learning loop with DFT calculations to achieve high accuracy [6]. However, it relied on global chemical descriptors—such as electrotopological-state (E-state) indices, cheminformatics descriptors, and Extended Topochemical Atom (ETA) descriptors—that offer limited interpretability and are not readily applicable to structure-based molecular design. Similarly, a frontier orbital prediction model [1] adopted Klekota-Roth fingerprints (KRFPs) [14]—originally developed for bioactivity prediction—as input features to model HOMO and LUMO energy levels. While predictive performance was strong, the domain relevance of the descriptors was limited, reducing their utility in interpreting results for organic electronic systems. These examples highlight the urgent need for a domain-specific fingerprint that bridges chemical intuition and machine learning in organic electronics, making it difficult to translate ML model predictions into actionable molecular design insights. Previous studies have attempted to construct molecular fingerprints using polymer units to analyze structural features that influence target properties such as power conversion efficiency (PCE) [15], and to design high-performing molecules based on these insights [7]. This highlights the need to develop domain-specific fingerprints that effectively capture the characteristic structural motifs of organic electronic materials. To address these limitations, we propose the Organic Electronic Fingerprint (OEFP), a structure-based, 5,363-bit fixed-length molecular fingerprint specifically designed for OLED and OPV materials (Fig. 1). OEFP encodes synthetically accessible, chemically meaningful, and optoelectronically relevant substructures derived from a curated dataset comprising 864 OLED molecules, approximately 2.3 million OPV candidates [16], and 6,865 chromophores [17, 18]. Using the Breaking of Retrosynthetically Interesting Chemical Substructures (BRICS) fragmentation algorithm [19], we extracted frequently observed fragments and developed an additional algorithm to identify unit rings from fused ring clusters, accounting for the importance of π-conjugation in optoelectronic performance. These structural motifs were then mapped to individual fingerprint bits, allowing for both predictive accuracy and interpretability. By bridging machine learning with chemical intuition, OEFP enables transparent feature attribution (e.g., via SHAP analysis [20]), facilitates rational design, and integrates seamlessly with generative workflows. This positions OEFP as a foundational tool for accelerating the discovery and optimization of high-performance organic electronic materials.

Dataset

OLED

First and foremost, for OLED materials—our primary area of interest—publicly available datasets containing widely used OLED compounds are scarce. To address this, we manually curated a comprehensive dataset with

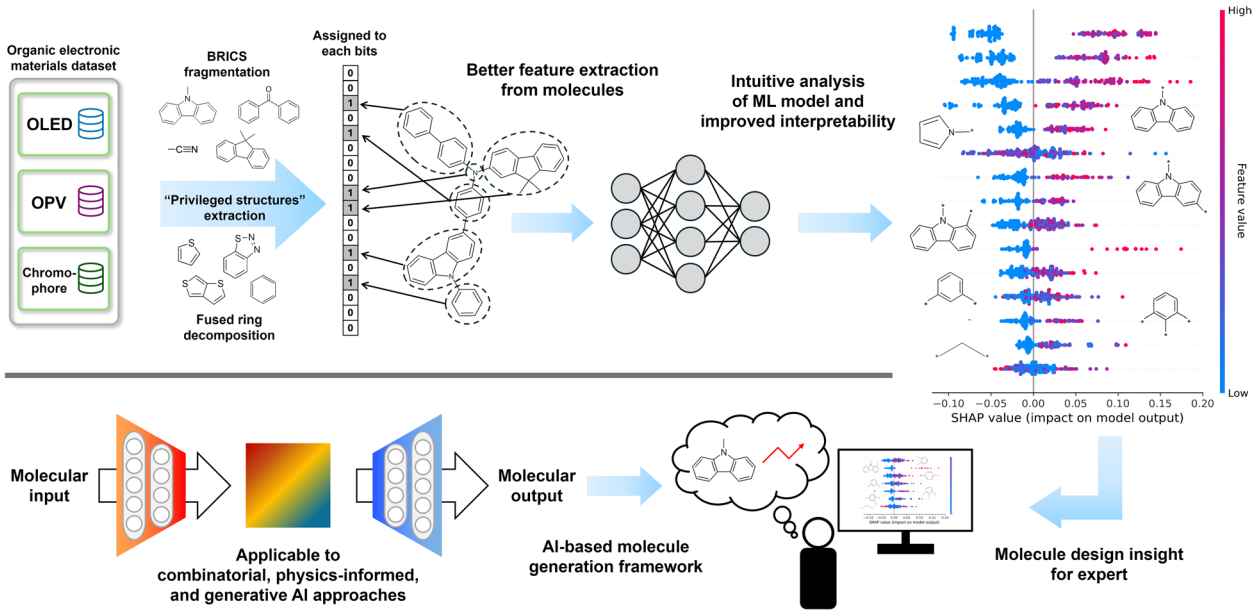


Fig. 1 Schematic workflow of OEFP construction and applications. Privileged substructures were derived using BRICS fragmentation and fused ring decomposition, then mapped to fingerprint bits. OEFP enables both accurate molecular property prediction and interpretable SHAP-based feature analysis, providing a foundation for AI-driven molecular design

864 molecules by extracting information from the literature. This includes widely used host [21] and dopant materials [22] across the red, green, and blue emission spectrum, as well as representative materials used in common layers such as ETL (Electron Transport Layer) [23, 24], EIL (Electron Injection Layer) [25], HTL (Hole Transport Layer) [26], and HIL (Hole Injection Layer) [27]. Furthermore, we incorporated TADF (thermally activated delayed fluorescence) materials, including both donor–acceptor (DA) types [28] and multi-resonance (MR) types, which have significantly contributed to improvements in OLED efficiency and lifetime. The curated molecular structures were standardized and converted into Simplified Molecular Input Line Entry System (SMILES) notation to enable consistent fragment extraction and efficient fingerprint generation. SMILES is widely used because it simplifies substructure matching and is compatible with most cheminformatics tools [29–31].

OPV

For OPV materials, we utilized the publicly available dataset from the Harvard Clean Energy Project Database (CEPDB) [16], one of the largest and most sophisticated computational screening efforts in organic electronics. The dataset contains approximately 2.3 million unique molecules generated through combinatorial assembly using 26 carefully selected molecular fragments. These fragments were linked and fused following chemical rules to yield small-molecule OPV candidates and oligomeric building blocks, creating a vast and diverse chemical space. What sets CEPDB apart is not only the scale of molecular coverage but also the rigorous computational strategy employed. The molecules were subjected to a multi-level screening pipeline involving quantum chemical calculations, cheminformatics descriptors, and property-based filtering. The CEPDB leveraged distributed computing via IBM's World Community Grid [32] to achieve this unprecedented scale. The molecular candidates were assessed for key photovoltaic properties such as HOMO–LUMO gap, open-circuit voltage (V_{OC}), and power conversion efficiency (PCE) using both first-principles calculations and machine learning models. Among these, high-efficiency candidates were prioritized for further study and database integration. The breadth, quality, and mechanistic relevance of this dataset make it well-suited for training and validating OEFP fingerprints in the context of OPV research. By applying OEFP to this dataset, we aim to extract domain-specific substructures with clear synthetic relevance and electronic function, ensuring the resulting fingerprint is robust, interpretable, and scalable across a wide range of OPV chemistries.

Chromophore

To further expand the dataset, we incorporated a comprehensive set of chromophore molecules reported by Joung et al. [17, 18], which added approximately 6,865

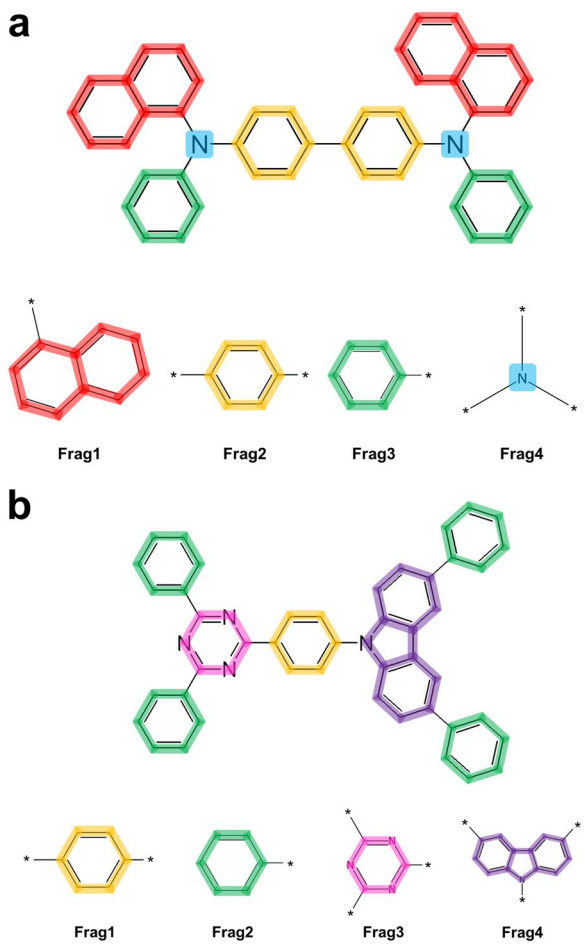


Fig. 2 Examples of BRICS-based fragmentation in OLED molecules. Molecular structures and their corresponding fragments of two OLED materials **a** N,N′-Di(1-naphthyl)-N,N′-diphenyl-(1,1′-biphenyl)-4,4′-diamine (NPB) and **b** 9-(4-(4,6-Diphenyl-1,3,5-triazin-2-yl)phenyl)-3,6-diphenyl-9H-carbazole (PCzTrz). This demonstrates how OLED molecules are decomposed into interpretable building blocks that can be directly mapped to OEFP bits

SMILES entries. These combined sources provide a robust and diverse representation of organic electronic molecular space [1].

Results and analysis

Fragmentation – BRICS algorithm

We employed the BRICS (Breaking of Retrosynthetically Interesting Chemical Substructures) algorithm [19] to fragment molecules into synthetically accessible units at chemically relevant disconnection points (Fig. 2a,b). This approach provides interpretable building blocks that reflect retrosynthetic logic and can be reassembled, ensuring both chemical relevance and synthetic feasibility.

To enhance the generalizability and reusability of resulting fragments, all BRICS-specific atom labels were replaced with a universal wildcard "[!#1]" during the SMARTS [33, 34] conversion process. This representation does not indicate the specific atom to which a bond was broken; rather, it conveys the common information that a bond was cleaved at that position. By abstracting away atom-specific details, the resulting fingerprint fragments avoid over-specialization and remain broadly applicable across diverse molecules. Following fragmentation, we compiled a frequency distribution of all generated fragments across the dataset and selected the top-N [14] most frequently occurring fragments. These fragments were then assigned to individual bits in the OEFP vector. This selection process ensures that OEFP captures the most structurally and functionally relevant motifs common to OLED and OPV molecules.

Importantly, the rationale behind using BRICS and selecting retrosynthetically accessible fragments extends beyond prediction tasks. OEFP was also designed with generative modeling in mind. When integrated with frameworks such as Variational Autoencoders (VAEs) [3, 4], the use of BRICS-derived substructures enables molecule generation that is not only chemically reasonable but also practically synthesizable. This aligns with the broader direction of data-driven de novo molecular design [35], where interpretability and synthetic accessibility are increasingly emphasized. This design choice is supported by prior work in drug discovery, where BRICS has been shown to outperform classical RECAP fragmentation [36] in reconstructing high-quality, 'drug-like' molecules with greater chemical coverage and diversity. As demonstrated in Degen et al., the BRICS framework offers a more sophisticated model of fragmentation and recombination and generates fragment spaces that preserve both synthetic feasibility and chemical complexity. By encoding OEFP with BRICS-based units, we simultaneously enhance model interpretability, synthetic relevance, and generative potential—key qualities for any molecular representation intended to bridge predictive modeling and chemical discovery.

Fragmentation–fused ring cluster decomposition

To complement BRICS-based fragmentation, we developed a ring decomposition algorithm to extract aromatic rings from fused ring clusters. BRICS alone could not fully capture the wide diversity of ring systems present in organic electronic materials, as complex fragments often emerged as combinations of multiple rings. By decomposing these fused clusters

into their constituent rings, our approach ensures a lossless representation of the fundamental ring motifs, enabling OEFP to better account for the structural features critical to OLED and OPV materials.

The method involves the following steps:

Molecular Graph Representation: Each molecule is represented as a molecular graph

$$G = (V, E) \tag{1}$$

where **V** is the set of atoms (vertices) and $\mathbf{E} \subseteq \mathbf{V} \times \mathbf{V}$ is the set of chemical bonds (edges).

Ring Detection: Rings

$$R = \{R_1, R_2, \ldots, R_n\} \tag{2}$$

are identified using RDKit's [37] ring perception algorithm. Each ring $\mathbf{R}_k \subseteq \mathbf{V}$ satisfies $\mathbf{5} \leq |\mathbf{R}_k| \leq \mathbf{7}$, focusing on 5-, 6-, and 7-membered aromatic rings. Fused Ring Clustering: Two rings $\mathbf{R}_i$ and $\mathbf{R}_j$ are considered fused if their intersection contains at least two atoms:

Adjacency Graph:

$$\left|R_i \cap R_j\right| = 2 \tag{3}$$

An adjacency graph is constructed based on this condition, and connected components are identified using depth-first search (DFS) [38]. Each connected component forms a fused ring cluster $\mathbf{C}_l = \bigcup \mathbf{R}_i$ for all i in the component.

Ring Substructure Selection Criteria: From each cluster C_l, substructures are selected based on k-ring combinations:

For $\mathbf{k = 1}$, all single rings within the cluster are included.

For $\mathbf{k = 2}$, pairs of rings are selected only if they are fused (share 2 atoms), and each ring must consist of 5–7 atoms.

Subgraph to SMILES Conversion: Selected ring-based atom indices are used to extract corresponding subgraphs:

$$E_{sub} = \{e_{ij} \in E \mid V_i, V_j \in \cup R_k\} \tag{4}$$

$$G_{sub} = (V_{sub}, E_{sub}) \tag{5}$$

This is constructed and converted to SMILES. These fused ring fragments are incorporated into OEFP to represent the electronic and structural cores frequently observed in organic semiconductors. This complementary approach enriches OEFP by embedding motifs that often govern the optoelectronic performance of the target materials.

Composition of OEFP

To construct the Organic Electronic Fingerprint (OEFP), we extracted molecular fragments using the BRICS algorithm from three representative datasets: OLED (462 fragments), OPV (605,747 fragments), and chromophore molecules (4,050 fragments). Our goal was to capture the structural diversity of organic electronic materials while keeping the fingerprint bit length comparable to existing fingerprints. Therefore, we selected top-ranked frequent fragments from each dataset while maintaining domain-specific proportional representation.

For the OLED dataset, which had the fewest fragments, we included all fragments with a frequency of 1 or higher to ensure adequate representation of molecular features (Fig. 3d). In the OPV dataset, we set the cutoff at a frequency of 32. This threshold captured 3,188 fragments—beyond which the fragment count increased sharply (e.g., 24,899 fragments at 31), rendering the fingerprint impractically long (Fig. 3e). To confirm that this frequency cutoff appropriately balances feature diversity and dimensional efficiency, we additionally compared predictive performance at two fragment counts (375 bits @ cutoff = 35 and 3,188 bits @ cutoff = 32) across both the CEPDB 10 k and National Renewable Energy Laboratory (NREL) OPV 81 k datasets (Table. S1) [39, 40]. The results showed that expanding the fragment coverage to 3,188 bits consistently reduced MAE and improved R^2, supporting our decision to adopt the 32 cutoff as an optimal balance between representational richness and computational tractability.

Similarly, for the chromophore dataset, we included fragments with a frequency of 2 or higher, resulting in 1175 fragments. Including single-occurrence fragments would have added 2,875 additional entries, again leading to excessive fingerprint length (Fig. 3f). In addition to BRICS fragments, we incorporated decomposed fused rings into OEFP—112, 73, and 641 ring fragments from the OLED, OPV, and chromophore datasets, respectively (Fig. 3b). Because ring fragments are fewer in number yet crucial for electronic properties, all were retained without frequency filtering. As a result of frequency-based selection, 602,559 OPV fragments and 2,875 chromophore fragments were excluded from OEFP. To ensure that the selected features still captured the key structural characteristics of the excluded low-frequency fragments, we performed a detailed atom-level coverage analysis. For each excluded fragment, we counted the number of non-hydrogen atoms and calculated the proportion of atoms that were matched by any of the 5,363 OEFP bits. Although described as atom-level matching, this effectively reflects subgraph-level correspondence, indicating how completely each functional moiety or structural motif is represented within OEFP.

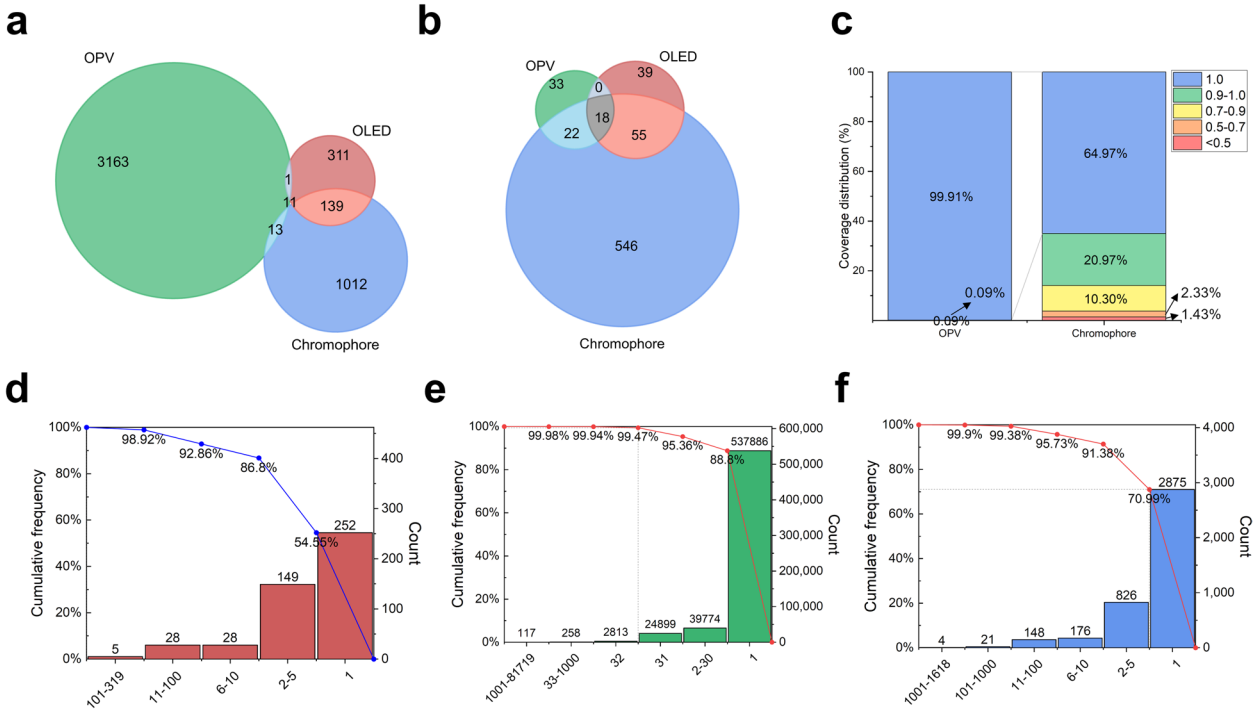


Fig. 3 Fragment selection and coverage analysis across OLED, OPV, and chromophore datasets. Distribution of top-N fragments extracted from each dataset using the **a,** BRICS fragmentation and **b,** fused ring cluster decomposition method, among those included in the OEFP. **c,** Coverage rate distribution of OPV and chromophore fragments not included in OEFP, measured by the proportion of atoms matched by OEFP bits. The resulting coverage rates were visualized using a color gradient, with blue representing fully matched fragments (100% coverage) and red indicating fragments with less than 50% matching. **d–f,** Frequency distributions and cumulative frequencies of BRICS fragments in OLED (**d**), OPV (**e**), and chromophore (**f**) datasets. Each histogram displays fragment frequency intervals on the x-axis. The left y-axis represents the cumulative frequency (%), while the right y-axis shows the number of fragments in each frequency bin. Together, these results confirm that OEFP achieves a compact representation while retaining comprehensive structural coverage, ensuring robustness and interpretability across diverse molecular datasets

As shown in Fig. 3c, for the OPV dataset, among 602,559 excluded fragments, 99.91% (602,006) were fully covered, and an additional 0.09% (553) exhibited coverage above 90%. For the more diverse chromophore dataset, 64.97% (1,868 of 2,875) of excluded fragments were fully covered and 85.94% (2,471) exceeded 90% coverage, while 98.57% (2,834) retained coverage over 50%. These results demonstrate that OEFP faithfully represents both frequent and rare structural motifs.

Together, the frequency-based optimization (validated in Table. S1) and the subgraph-level coverage analysis confirm that the 32 cutoff yields a compact yet chemically comprehensive fingerprint. To further examine whether a purely frequency-based fragment selection may discard rare but informative motifs, we performed an additional analysis on the NREL OPV dataset. Since the OPV fingerprint construction applies a frequency cutoff of 32, we specifically investigated whether fragments below this threshold contain low-frequency yet high-informativeness motifs that could materially affect model performance. We evaluated fragments with low frequencies using simple univariate association metrics between fragment presence and the target property, including the absolute difference in conditional means, absolute t-statistics, and effect-size–like correlation measures. This analysis was designed to identify fragments that, despite low occurrence, show a strong statistical association with the target. Based on this criterion, we identified a small subset of 22 fragments exhibiting relatively high informativeness scores. Notably, these fragments were not extremely rare. All selected motifs fall within a narrow frequency range (approximately 29–31 occurrences), and no fragments with substantially lower frequencies showed strong univariate associations. We then augmented the original OEFP representation with these 22 fragments and systematically evaluated the impact on predictive performance under both random and scaffold-based splits.

The results show that incorporating these fragments does not lead to a consistent improvement in model performance. Under random splits (Fig. S10a), prediction errors increased and R^2 decreased, while under scaffold

splits (Fig. S10b), only marginal and non-systematic changes were observed. Overall, the performance variations were small and dependent on the splitting strategy, indicating no robust benefit from explicitly including these low-frequency fragments. These results suggest that, for the OPV datasets considered here, the number of rare yet highly informative fragments below the frequency cutoff is limited, and their inclusion does not materially alter predictive performance. Accordingly, frequency-based fragment selection offers a practical balance between coverage and dimensional efficiency, while preserving the essential chemical features required for robust and interpretable machine-learning-based property prediction.

Validation via machine learning models — OPV

In machine learning for molecular property prediction, the choice of molecular descriptors and the size of the training dataset critically affect model performance. To assess the robustness and generalizability of OEFP across different data regimes, we randomly extracted subsamples of 1 k and 10 k molecules from the NREL OPV database and utilized the full 81 k training dataset. This design enabled a consistent evaluation of OEFP under varying data scales.

Before reporting model performance on the NREL benchmark, we assessed whether NREL can be treated as an external evaluation set relative to CEPDB. An exact SMILES match between the full CEPDB and NREL databases found only four duplicated molecules, suggesting negligible explicit leakage from identical structures. We further quantified molecular-level similarity using Morgan fingerprints (4,096 bits, radius = 2) and Tanimoto similarity. Because an exhaustive CEPDB–NREL all-vs-all comparison is computationally infeasible, we compared 81 k NREL molecules against a random subset of 200 k CEPDB molecules and recorded, for each NREL molecule, the maximum similarity value (Table. S2). The similarity distribution is centered at low values (mean 0.26; median 0.25), with p95 and p99 of 0.39 and 0.47, respectively, and fewer than 1% of NREL molecules showing a maximum similarity above 0.5. Together, these results indicate that most NREL molecules are only moderately similar to the CEPDB corpus at the molecular level, supporting the use of NREL as a practically external benchmark despite partial overlap in scaffold space.

For comparison, in addition to binary and count-based OEFP, we included KRFP (4,860 bits; privileged substructure-based, interpretable but not domain-specific) and ECFP4/6 (Extended-Connectivity Fingerprints with radii of 2 and 3, respectively), which capture general atomic environments irrespective of material domain. The bit length of 4096 was selected to maintain dimensional consistency with OEFP (5,363 bits) and KRFP (4,860 bits). All fingerprints were evaluated using an identical XGBoost regression framework, optimized via Bayesian optimization for six hyperparameters, ensuring a fair comparison of descriptive performance.

To test scalability, models were trained on datasets containing 1 k, 10 k, and 81 k molecules. The 1 k datasets (five random samples) were evaluated via fivefold cross-validation for statistical robustness, whereas the 10 k datasets (five random samples) and the 81 k dataset used an 8:2 train–test split shared across all fingerprints. For the full dataset, bootstrap resampling (1,000 samples × 5,000 iterations) was applied to the 16,000 test molecules to estimate the 95% confidence interval (Table. S3, Fig. S1-S8). An Message Passing Neural Network (MPNN) [41, 42] model was additionally trained on the 81 k dataset to serve as a deep-learning baseline [43, 44].

Across all dataset sizes, OEFP exhibited consistent superiority over KRFP and competitive performance with ECFP (Table 1).

For the smallest dataset (1 k molecules) (Fig. 4a, d), OEFP reduced MAE by 4–5% relative to KRFP and achieved comparable R^2 gains. Although ECFP4/6 achieved slightly lower MAE (0.139–0.120 eV), OEFP (count) retained competitive accuracy (0.141 eV) while offering domain-specific interpretability.

In the 10 k datasets (Fig 4b, e), OEFP again outperformed KRFP (ΔMAE = 1.3–9.1%, $\Delta R^2 = +0.9$–1.6%) and remained close to ECFP (MAE = 0.091 eV vs 0.086–0.089 eV), demonstrating that OEFP preserves chemical relevance even with fewer bits.

For the 81 k dataset (Fig 4c, f), OEFP (count) maintained its lead over KRFP (count) with a 13% lower MAE (0.051 vs 0.059 eV) and higher R^2 (0.971 vs 0.961). While ECFP yielded marginally better values (MAE ≈ 0.041 eV, $R^2 \approx 0.982$), OEFP (count) achieved 96–98% of ECFP's predictive performance with substantially higher interpretability.

The graph-based MPNN achieved MAE = 0.037 eV, $R^2 = 0.987$, validating its expected superiority in expressive power. When OEFP was used in the binary format, its performance lagged behind MPNN by 0.08 (MAE) and 0.112 (R^2), whereas in the count format, the gap narrowed to only 0.014 eV (MAE) and 0.016 (R^2). This indicates that count encoding enables OEFP to approach deep-learning accuracy while preserving interpretability and low computational cost.

Model performance on the NREL OPV dataset is primarily discussed using random data splits to enable direct comparison with prior studies. Under a random 8:2 split, Bemis–Murcko scaffold analysis revealed substantial overlap between the training and test sets, with 4,571 shared scaffolds out of 5,157 test scaffolds,

Table 1. Performance comparison of molecular fingerprints on the OPV dataset

Fingerprint	1 k dataset		10 k dataset		81 k dataset	
	MAE (eV)	R^2	MAE (eV)	R^2	MAE (eV)	R^2
OEFP binary	0.1820 ± 0.010	0.6856 ± 0.047	0.1406 ± 0.004	0.8237 ± 0.013	0.1182	0.8747
OEFP count	0.1408 ± 0.011	0.8089 ± 0.043	0.0911 ± 0.004	0.9169 ± 0.013	0.0510	0.9709
KRFP binary	0.1903 ± 0.015	0.6826 ± 0.038	0.1424 ± 0.002	0.8141 ± 0.010	0.1234	0.8667
KRFP count	0.1473 ± 0.011	0.8043 ± 0.037	0.1002 ± 0.003	0.9021 ± 0.007	0.0591	0.9614
ECFP4 binary	0.1387 ± 0.009	0.8061 ± 0.039	0.0863 ± 0.003	0.9249 ± 0.009	0.0510	0.9723
ECFP4 count	0.1185 ± 0.008	0.8605 ± 0.037	0.0721 ± 0.005	0.9454 ± 0.009	0.0414	0.9823
ECFP6 binary	0.1467 ± 0.009	0.7927 ± 0.036	0.0889 ± 0.002	0.9199 ± 0.004	0.0433	0.9788
ECFP6 count	0.1196 ± 0.007	0.8543 ± 0.023	0.0734 ± 0.004	0.9439 ± 0.008	0.0405	0.9811
MPNN	–	–	–	–	0.0372	0.9866

Model performance for OEFP, KRFP, and ECFP fingerprints trained on the OPV datasets of different sizes (1 k, 10 k, and 81 k molecules), evaluated using XGBoost regression. Each fingerprint was tested in both binary and count-based formats. For the 1 k datasets, reported values correspond to the median ± IQR/2 (from 25th–75th percentile) computed across five independently sampled datasets, each evaluated by five-fold cross-validation (25 evaluations in total). For the 10 k datasets, values were derived from five independent datasets, each using a single 8:2 train–test split. For the 81 k dataset, a single 8:2 split was used, and the statistics reflect 5,000 bootstrap resamples of the 16,000 test molecules. Performance metrics include MAE, RMSE, R^2, and Pearson's r, which collectively assess prediction accuracy and model consistency.

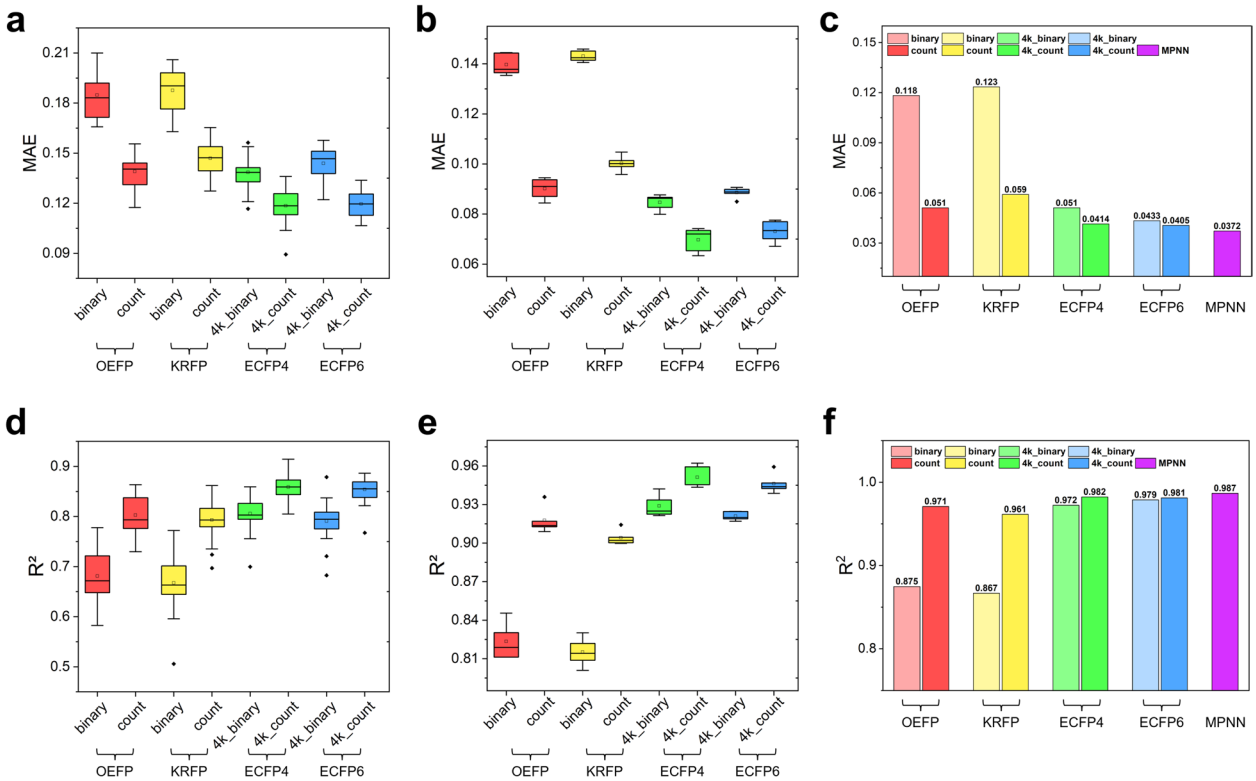


Fig. 4 Performance comparison of models trained on subsampled NREL organic photovoltaic database using different fingerprint types and sizes. Test set results from models trained on randomly subsampled NREL OPV DB of 1 k (**a, d**). 10 k (**b, e**), and the entire 81 k molecules (**c, f**). For 1 k, 10 k dataset size, five independent datasets were randomly sampled. Each of these datasets was then split into training and test sets using an 8:2 ratio. Panels **a–c** show the distribution of MAE, and panels **d–f** show the corresponding R^2 scores. Each model was evaluated using OEFP (binary/count), KRFP (binary/count) and ECFP with radii of 2, 3 and bit length of 4,096. For the full 81 k dataset, an MPNN model was additionally trained to provide a stronger baseline for comparison

corresponding to an overlap ratio of 0.89 and a Jaccard index of 0.48. To assess robustness under scaffold shifts, we therefore additionally performed a scaffold-based evaluation using an 8:1:1 scaffold split with no scaffold overlap, with the results summarized in Fig. S9. Under this scaffold-based split, performance differences became more pronounced. OEFP retained reasonable accuracy (MAE = 0.132/0.078 and R^2 = 0.831/0.925 for binary/count), whereas KRFP showed a severe degradation, while ECFP4/6 showed slightly higher predictive performance. Notably, although the OEFP–ECFP gap increased relative to random splits, the OEFP–KRFP gap widened much more substantially, indicating that OEFP generalizes more robustly to unseen scaffolds. We attribute this behavior to the fact that OEFP is constructed from OPV-relevant substructures mined from the target domain, enabling transfer across different scaffold topologies.

Collectively, these findings confirm that OEFP provides a balanced fingerprint representation, combining predictive accuracy, structural interpretability, and scalability, particularly effective for organic electronic materials, where repeated or symmetric motifs make count-based encoding crucial for accurate property prediction. By design, OEFP is a substructure-based molecular representation that emphasizes local chemical environments, including heteroatom identity and position, short-range ring connectivity, and local bonding motifs. Accordingly, the OLED, OPV, and chromophore datasets considered in this work fall within the intended scope of OEFP, as their electronic properties are largely governed by such local structural features, whereas effects arising purely from very long-range conjugation or extended fused-ring delocalization are not explicitly encoded.

Model interpretability via SHAP analysis — OLED

To investigate which specific substructures significantly influence the material properties of OLED molecules and to validate the interpretability of OEFP in molecular machine learning, we constructed predictive models using a dataset of 864 OLED molecules, each annotated with its HOMO energy level measured via cyclic voltammetry. The molecular structures were encoded into count-based OEFP vectors, selected based on their superior predictive performance in the OPV dataset. Using five-fold cross-validation and Bayesian optimization, the XGBoost model achieved a MAE of 0.184 eV and an R^2 of 0.686, outperforming KRFP (MAE = 0.191 eV, R^2 = 0.559) under identical conditions. Although the hash-based ECFP occasionally exhibited higher predictive performance than OEFP in the previous OPV benchmarks, OEFP provides a unique advantage that ECFP fundamentally lacks direct interpretability (Fig. 4)

While ECFP encodes atomic environments via hashing, its bit indices cannot be directly associated with specific chemical substructures. In contrast, OEFP assigns fixed indices to chemically meaningful fragments, enabling the direct attribution of predicted properties to individual functional groups. This interpretability is particularly valuable in OLED materials, where most compounds share a limited set of recurring donor–acceptor motifs. The OLED-specific substructure dictionary of OEFP therefore allows clearer mechanistic understanding than the biologically oriented KRFP.

To further assess this interpretability, we performed SHAP analysis on the count-based OEFP and KRFP models to quantify and visualize the contribution of each substructure to HOMO energy level predictions. The analysis successfully identified the top fragments most strongly influencing the HOMO level (Fig. 5).

For example, OEFP160, corresponding to carbazole, a well-known electron-donating fragment, exhibited a strong positive correlation with HOMO values: higher carbazole frequency led to elevated HOMO levels, whereas its absence resulted in lower values. In contrast, OEFP8, corresponding to phenyl, an electronically neutral group, showed no consistent correlation, consistent with chemical intuition.

A more detailed comparison revealed that donor-type fragments critical to OLED electronic properties are largely absent from KRFP. The KRFP dictionary—derived mainly from bioactive substructures—does not explicitly include carbazole or related donor motifs. Instead, KRFP fragments such as KRFP3083, 2071, and 2068 correspond only to partially matching phenyl-carbazole-like patterns.

Conversely, OEFP clearly captured chemically meaningful donor fragments: OEFP3279, 161, and 140, all strong electron-donating groups, ranked among the most influential features, along with multiple carbazole-derived motifs such as OEFP160, 372, and 4080. Although the SHAP interpretation here was limited to the top 15 fragments, these results demonstrate that OEFP successfully encodes a broad range of chemically interpretable donor and π-conjugated units relevant to OLED performance. Such correspondence between structural motifs and electronic properties provides valuable design insight that is not readily accessible in a direct and domain-aligned manner using traditional hashed fingerprints such as ECFP or domain-mismatched KRFP. While KRFP is interpretable in principle, its predefined motifs are not tailored to the OPV/OLED domain and therefore capture only limited or overly coarse structural information. In contrast, interpretation of ECFP typically requires additional post hoc procedures to reconstruct atom-centered environments from hashed representations. By explicitly defining a library of domain-relevant

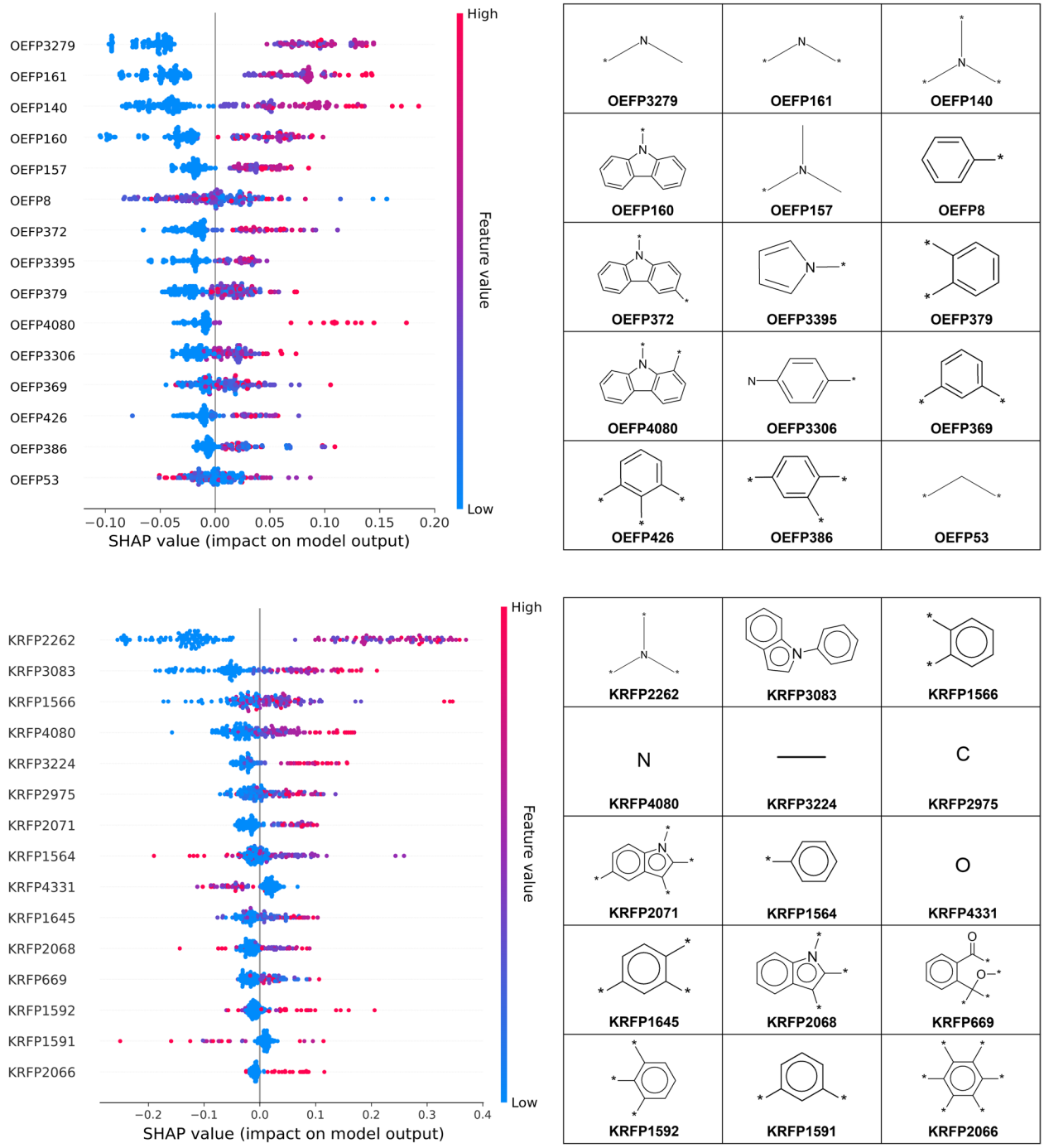


Fig. 5 SHAP analysis of OEFP and KRFP features and corresponding substructures. SHAP summary plot showing the 15 OEFP and KRFP bits with the highest impact on HOMO energy level prediction of OLED materials. The left panel displays the SHAP values for each feature, color-coded by feature value. The right panel visualizes the molecular substructures matched to each bit. We used a count-based representation of OEFP and KRFP, where each bit reflects the number of occurrences of the corresponding substructure, rather than a simple binary presence or absence. This analysis demonstrates that OEFP provides chemically intuitive and interpretable structure–property relationships

fragments, OEFP enables immediate, fragment-level interpretation without additional processing, allowing even non-expert users to intuitively relate model predictions to chemically meaningful substructures and to leverage this insight for rational molecular design. Although the SHAP interpretation here was limited

to the top 15 fragments, these results demonstrate that OEFP successfully encodes a broad range of chemically interpretable donor and π-conjugated units relevant to OLED performance.

Conclusion

The Organic Electronic Fingerprint (OEFP) provides a domain-specific representation tailored for organic electronic materials, simultaneously achieving predictive accuracy and structural interpretability, thereby establishing its academic significance. Empirical validation across OPV datasets of varying sizes demonstrated that OEFP achieved up to 13.7% lower MAE and 1.6% higher R^2 compared to conventional fingerprint such as KRFP, while an OLED case study confirmed its ability to provide intuitive interpretation of structure–property relationships through SHAP analysis, yielding direct insights into molecular design. At the same time, because OEFP is constructed from synthetically accessible substructures, it offers practical extensibility toward computer- and AI-assisted molecular design, providing a pathway to accelerated discovery of high-performance OLED and OPV materials.

Methods

Mean absolute error (MAE)

MAE [45] measures the average magnitude of errors by calculating the average absolute difference between actual and predicted values. Unlike mean squared error (MSE) [46] or root mean squared error (RMSE) [47], which square the errors and therefore emphasize larger deviations, MAE evaluates prediction errors linearly, making it easier to intuitively interpret how far, on average, the predictions deviate from the actual values. This makes MAE particularly useful when a straightforward assessment of model accuracy is desired. The formula for MAE is:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_l\right| \tag{6}$$

where, n is the number of samples, y_i is the actual value, $\widehat{y_i}$ is the predicted value.

R^2 score

The R^2 score [48], also known as the coefficient of determination, is defined as:

$$R^2 = 1 - \frac{\sum_{i-1}^{n}\left(y_i - \hat{y}_l\right)^2}{\sum_{i-1}^{n}\left(y_i - \bar{y}\right)^2} \tag{7}$$

Where n is the total number of samples, y_i represents the actual value for the i-th sample, $\hat{y}_i$ is the predicted value for the i-th sample, and $\bar{y}$ is the mean of the actual values. The numerator, $\Sigma\ (y_i—\hat{y}_i)^2$, is called the residual sum of squares (RSS), which measures the total squared error between predictions and actual values. The denominator, $\Sigma\ (y_i—\bar{y})^2$, is the total sum of squares (TSS), representing the total variance in the actual values. R^2 quantifies how well the model explains the variance in the outcome variable. An R^2 of 1 indicates perfect prediction, meaning all predicted values exactly match the actual values. An R^2 of 0 implies that the model does no better than simply predicting the mean of the actual values for all observations.

SHAP method

The SHapley Additive exPlanations (SHAP) [20] is a unified framework for interpreting machine learning model predictions based on Shapley values from cooperative game theory. It quantifies the contribution of each feature to the final prediction by fairly distributing the prediction difference across all features.

$$\Phi_i = \sum\nolimits_{S\subseteq N\setminus\{i\}} \frac{|S|\,!(|N| - |S| - 1)\,!}{|N|\,!} \tag{8}$$

The SHAP value $\boldsymbol{\phi_i}$ represents the average marginal contribution of feature i to the model's prediction across all possible subsets S of the remaining features (excluding i). Specifically, for each subset S, the model's prediction is computed both with and without feature i, and the difference $\mathbf{f(S\cup\{i\})—f(S)}$ indicates how much adding feature i changes the output. This difference is then weighted by the term $\mathbf{(|S|!\times(M—|S|—1)!)\ /\ M!}$, where **M** is the total number of features and $\mathbf{|S|}$ is the size of the subset S. The weighted contributions across all subsets are then summed to produce the SHAP value $\boldsymbol{\phi_i}$.

Message passing neural network

To evaluate the structure-based predictive performance of graph neural networks, we trained a message-passing neural network (MPNN) implemented with the PyTorch Geometric framework (v2.5). All molecules were converted from SMILES strings to molecular graphs, where each atom was represented by a 4-dimensional feature vector [Z, degree, formal charge, aromaticity], and each covalent bond was described by its bond-order type. A Graph Isomorphism Network (GIN) architecture was adopted to perform neighborhood aggregation, consisting of three GINConv layers with learnable multi-layer perceptrons (MLPs), followed by a global mean-pooling operation to generate graph-level embeddings. The pooled embeddings were processed through two fully connected layers ($128\rightarrow64\rightarrow1$) with ReLU activation

and a dropout rate of 0.1. The dataset (81,000 OPV molecules) was randomly divided into training, validation, and test subsets with an 8:1:1 ratio. Training was performed using the Adam optimizer (learning rate = 1×10^{-3}, batch size = 64) for 300 epochs with an L1 loss function. Model performance was evaluated using mean absolute error (MAE), root-mean-square error (RMSE), coefficient of determination (R^2), and Pearson's correlation coefficient (r). The model achieving the lowest validation MAE was saved and used for final test evaluation. All experiments were conducted with a fixed random seed (42) to ensure reproducibility. The complete training and inference pipeline, including SMILES-to-graph conversion, data splitting, model configuration, and checkpoint saving, was implemented in Python (PyTorch 2.8.0, RDKit 2023.09, CUDA-enabled GPU).

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13321-026-01169-7.

Supplementary Material 1.

Acknowledgements
This work was supported by grants from the Industrial Strategic Technology Development Program (RS-2024-00418116) and Technology Innovation Program (RS-2025-25452683) through Korea Planning & Evaluation Institute of Industrial Technology (KEIT), funded by the Ministry of Trade, Industry and Resources, Korea. It was also supported by Basic Science Research Program through the NRF, by the Ministry of Science and ICT, Korea (2021R1A2C1008725).

Author contributions
T.W.Y. contributed to the conception of the study. T.W.Y., S.H.J., and M.C.S. contributed to the design of the work. T.W.Y. and S.H.J. contributed to data acquisition and analysis. T.W.Y. contributed to data interpretation and the creation of new software. T.W.Y. drafted the manuscript. T.W.Y., S.H.J., and M.C.S. edited and revised the manuscript. M.C.S. supervised the project, provided funding acquisition, project administration, and resources, and served as the corresponding author. All authors reviewed and approved the manuscript.

Funding
This work was supported by grants from the Industrial Strategic Technology Development Program (RS-2024–00418116) and Technology Innovation Program (RS-2025-25452683) through Korea Planning & Evaluation Institute of Industrial Technology (KEIT), funded by the Ministry of Trade, Industry and Resources, Korea. It was also supported by Basic Science Research Program through the NRF, by the Ministry of Science and ICT, Korea (2021R1A2C1008725).

Data availability
OEFP, including the bit-assigned substructure list and related source code, is available under the open-source MIT License at https://github.com/OLEDman926/OEFingerprint to ensure reproducibility and facilitate further research.

Declarations

Competing interests
The authors declare no competing interests.

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Received: 30 August 2025 Accepted: 13 February 2026
Published online: 27 February 2026

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